# Decarbonizing Heat: The Impact of Heat Pumps and a Time-of-Use Heat Pump Tariff on Energy Demand*

Louise Bernard
*Centre for Net Zero*

Andy Hackett
*Centre for Net Zero*

Robert Metcalfe
*Columbia University*
*Centre for Net Zero & NBER*

Andrew R. Schein
*Centre for Net Zero*
*London School of Economics*

October 14, 2026

**Abstract**: Heat pumps have been proposed as the leading technology in the electrification of domestic heat and therefore could play a crucial part in decarbonizing energy systems. However, there is limited causal evidence on how they affect energy demand and how energy prices can optimize their use. Leveraging a staggered rollout of heat pumps by a large energy supplier in Great Britain, we show: (1) heat pumps reduce average household gas use by 98% and increase electricity use by 50%, leading to a 72% reduction in carbon emissions from household energy consumption over a typical asset lifetime; (2) a time-of-use tariff (a retail electricity pricing plan) designed for heat pumps can halve electricity demand during the evening peak to enhance grid flexibility; (3) the marginal value of public funds for the current heat pump subsidy in Great Britain is £1.23 per £1 of government spending. Overall, we find that heat pumps can effectively decarbonize heat and subsidies can be welfare-enhancing.

> *"The reversed-cycle heat engine (or heat pump) is described, and it is shown... that where heat at a comparatively low temperature is required, heating efficiencies of the order of from 300 to 500 per cent or even more may be obtained."*
>
> — Graeme Haldane, *1930, p. 666.*

*Bernard: Centre for Net Zero (email: louise.bernard@centrefornetzero.org). Hackett: Centre for Net Zero (email: andy.hackett@centrefornetzero.org). Metcalfe: Columbia University, Centre for Net Zero, and NBER (email: rdm2176@columbia.edu). Schein: Centre for Net Zero and London School of Economics (email: andrew.schein@centrefornetzero.org). The order of authors is alphabetical.

**Acknowledgements**: We are very grateful to Lucy Bourke, Daniel Lopez Garcia, Izzy Woolgar, and Lucy Yu for their comments and formative critique. We thank Cal Gibson at Octopus Energy, who was a fundamental partner in our processing and understanding of Octopus Energy's consumer adoption and consumption data. We also thank Alex Schoch, Geraldine De Boisse, Mat Oates, Ben Ferris, Jaye Cribb and many others at Octopus Energy Group for implementing the heat pump program and *Cosy Octopus* heat pump time-of-use tariff, working with us to survey Octopus Energy households about how they use their heat pumps, and helping us process and use Octopus Energy household data. We are thankful to Oli Berry, Edwin Chan, Jess Colleen, Lucas Davis, Nathan Hendren, Chris Knittel, Sethu Odayappan, Toby Park, Andy Regan, TJ Root, Andrew Sissons, Blake Shaffer, Phil Steele, Kieron Stopforth, and Max Woollard for excellent comments. We thank Navya Kumar and Chloe Lucia Nemo Ramirez for excellent research assistance. All mistakes herein are ours alone. Conflicts of Interest: Centre for Net Zero is part of Octopus Energy Group Limited.

# 1 Introduction

Heating buildings accounts for nearly one-fifth of global energy-related carbon emissions, much of which comes from residential gas boilers in industrialized countries (International Energy Agency 2023). Replacing boilers with electric heat pumps powered by low-carbon electricity is a key element of decarbonization strategies recommended by international institutions such as the International Energy Agency (2021) and IPCC (2022).

Heat pumps have long been recognized for their potential engineering energy efficiency advantage in converting energy into useful indoor heat. Engineering analyses dating back to Haldane (1930) highlight that, relative to electric resistance heating, heat pumps can deliver about three times as much heat per unit of energy consumed. At the same time, residential heating in the UK remains overwhelmingly reliant on gas, with the vast majority of households using gas boilers as the incumbent technology. While gas and electricity have similar marginal carbon intensities in the UK today, energy efficiency gains from switching from gas boilers to electric heat pumps have the potential to substantially reduce carbon emissions. This reduction in emissions will increase further if, as projected, the marginal carbon intensity of electricity declines as the grid decarbonizes. However, there remains little causal evidence on the realized performance of heat pumps or on how their widespread use could affect energy systems. Empirical evidence is crucial for electricity system operators and for many governments around the world considering or using a heat pump subsidy.

This paper provides the first large-scale evidence on how heat pumps affect household energy demand, using detailed consumption data from a large utility in Great Britain. We exploit a staggered rollout of installations driven by supply constraints to identify impacts using a difference-in-differences design (Callaway and Sant'Anna 2021). Our setting offers rare conditions for identification: all households in the sample eventually receive a heat pump, but adoption occurs at different times for reasons unrelated to energy use.

Our contribution is threefold. First, from a sample of 1,308 households, we find that adopting a heat pump causes an 8,744 kWh (98%) reduction in household gas consumption and a 2,551 kWh (50%) increase in electricity use. These impacts are in the context of a policy regime mandating removal of households' previous heating system, such as their gas boiler, in order to access the heat pump subsidy. The electricity and gas results together imply that real-world heat pump energy efficiency meets engineering predictions. We find lower-than-expected energy efficiency at colder temperatures and higher-than-expected energy efficiency at warmer temperatures; these may reflect behavioral or operational factors not captured in engineering models and represent the largest divergences between our empirical results and model-based predictions. Overall, we believe these results are the first credible validation that the theoretical energy efficiency gains of heat pumps translate into realized energy and carbon savings at scale.

The close correspondence between our estimates and engineering predictions suggests that

households maintained similar heating practices while switching fuels, rather than using the heat pump's greater efficiency to raise comfort levels. We do not observe indoor temperatures directly, so we infer the absence of such a rebound from the consumption data itself: if households had responded to the heat pump's lower marginal cost of comfort by raising indoor temperatures or heating for longer, realized energy savings would have fallen short of the engineering-based predictions. This is perhaps surprising given previous studies of the rebound effect (Allcott and Greenstone 2017; Brandon et al. 2022; Christensen et al. 2023; Fowlie et al. 2018). We believe two features of our setting help explain it: heat pump adoption here operates primarily through mechanical rather than behavioral channels, and operating costs remained similar after adoption, as electricity prices were approximately four times higher per unit of energy than gas prices during our analysis period.

This near-parity in operating costs, despite heat pumps' higher energy efficiency, largely reflects broader mispricing across fuels. Retail electricity prices embed a disproportionate share of fixed network and policy costs relative to gas, while gas prices exclude climate externalities. We show that correcting these distortions can lead heat pumps to deliver meaningful savings on operating costs.

Second, we show that time-of-use tariffs can materially shift demand, offering a pathway for cost-effective grid integration of electrified heat. Such tariffs are designed to shift electricity consumption away from times of grid constraints and higher wholesale energy costs. From a sample of 6,631 households, we analyze the introduction of a time-of-use tariff designed specifically for heat pump users. Heat pump adopters not facing a time-of-use tariff increase electricity consumption in all hours of the day, with higher increases in the morning. Adoption of the time-of-use tariff halves peak-period electricity consumption and doubles off-peak use. We see this load shifting even on the coldest days of the year and from all types of homes. These results indicate that, although heat pumps substantially shift heating demand onto the electricity grid, where marginal social costs fluctuate over time, time-of-use pricing can incentivize households to adjust the timing of consumption in ways that reduce congestion and exposure to high-cost periods. We also find that time-of-use pricing can reduce households' private operating costs substantially.

Third, we present evidence that the UK's heat pump subsidy (the Boiler Upgrade Scheme) is welfare-enhancing. Using our empirical estimates of the heat pump's impact on electricity and gas consumption, we evaluate the welfare implications of the subsidy based on computing its marginal value of public funds (Hahn et al. 2026; Hendren and Sprung-Keyser 2020). This calculation requires several assumptions, including the additionality of subsidy recipients (i.e., the share who would not have adopted a heat pump absent the subsidy) and the value placed on avoided carbon emissions. We discuss these, along with their sensitivity, in Section 3. Subject to these assumptions, our calculations suggest that the subsidy is welfare-enhancing. The main welfare benefit is a reduction of around 39% in carbon emissions from household energy consumption in 2024, rising to approximately 72% over an assumed 20-year lifetime of the heat pump as the electricity

grid decarbonizes (UK Government 2024). Under our preferred assumptions, we estimate that the heat pump subsidy generates roughly £1.23 in societal benefits per £1 in net government cost. That measure is the average marginal value of public funds (MVPF) across the £7,500 subsidy; because the MVPF declines with subsidy size, the first £1 of subsidy returns £1.65, rising to £2.51 if we include estimated learning-by-doing benefits.

It is important to note that our analysis takes place within the specific policy and technological context of the United Kingdom. In the UK, heat pump adoption has largely occurred through air-to-water systems designed to work with existing water-based radiators, and all of the heat pumps in our sample are of this type. These systems typically replace a household's gas boiler entirely. In many other settings, including much of continental Europe and the United States, air-to-air heat pumps are more prevalent. Air-to-air systems can also provide space cooling during the summer, a function that air-to-water systems generally do not offer. This difference is reflected in some of our results. For instance, the absence of increased electricity consumption during summer months reflects the fact that the systems in our sample do not provide cooling. In addition, the subsidy program we use for identification requires that the heat pump replace the household's previous heating system, whereas in other contexts heat pumps may be installed alongside existing heating technologies rather than fully displacing them.

Beyond these contextual and technological differences, our samples are not necessarily representative of the broader population of British households that have not yet adopted a heat pump or the heat pump tariff. Heat pump adopters in our study are likely those for whom heat pumps are most desirable or best suited, so the observed effects here need not reflect population treatment effects. We expect our energy consumption results to translate reasonably well beyond this sample, however, to the extent that the response we estimate is largely mechanical, driven by outdoor temperature and house size (Section 2.1.3), rather than by idiosyncratic characteristics of early adopters. The issue is more consequential for the heat pump tariff, where selection into treatment by structural winners or low-shifting-cost households is likely: our sample skews toward larger homes and technologically engaged households well placed to shift load at low cost, which likely explains why our estimated demand response (Section 2.2.2) is large relative to the time-of-use pricing literature. As tariff adoption extends beyond early adopters, who may be more likely to be structural winners, to the early majority, who may be more likely to have smaller and less automated homes, we expect smaller absolute load-shifting effects, although still meaningful ones relative to heat pump and home size.

Our findings relate to the energy economics literature on household responses to technological change and energy efficiency investments (Allcott and Greenstone 2017; Brandon et al. 2022; Christensen et al. 2023; Fowlie et al. 2018). In contrast to much of this literature, where realized energy savings from efficiency investments fall short of ex ante predictions due to behavioral responses, installation quality, or rebound, we find that heat pump adoption delivers reductions in fossil fuel use that are closely aligned with engineering expectations. The close alignment may stem, in part,

from the installation requirements attached to the heat pump subsidy. These include minimum design and commissioning standards, as well as the removal of the existing heating system, limiting the scope for under-performance driven by household or installer behavior. This agreement is reassuring from a policy perspective. Many decarbonization pathways rely heavily on modeled projections of heat pump performance. Our results suggest that, at least in this setting, such projections may be more reliable than those for other energy efficiency investments.

Our analysis also situates heat pump adoption within broader questions of policy effectiveness, household behavior, and energy system impacts. We extend work on the welfare analysis of energy subsidies (Hahn et al. 2026; Hendren and Sprung-Keyser 2020) and on household responses to time-varying electricity prices (Borenstein 2007; Imelda et al. 2024; Ito et al. 2023). Finally, we complement recent research on the determinants of heat pump adoption and diffusion (Anderson and Kirkpatrick 2024; Davis 2024), showing how adoption translates into concrete changes in household energy demand and emissions.

The remainder of the paper proceeds as follows. Section 2 presents the empirical strategy and results for the impact of heat pumps and the time-of-use tariff. Section 3 evaluates welfare implications, and Section 4 concludes. Full robustness checks, heterogeneity analyses, and methodological details are reported in the Supplemental Appendix.

## 2 Empirical Analysis

Our empirical analysis focuses on two related channels through which heat pump adoption affects energy use and system outcomes. First, we study how installing a heat pump alters households' energy consumption, focusing on the substitution from gas to electricity and the resulting implications for heating efficiency. Second, we examine how electrified heating interacts with retail electricity pricing by analyzing the effects of a time-of-use tariff designed for heat pump users. This tariff varies prices across the day to encourage load shifting away from high-cost periods, allowing us to assess whether time-varying prices can reallocate heat-pump demand within the day, mitigating peak load and system congestion, without materially changing average energy consumption.

Table 1 reports summary statistics for the electricity and gas consumption data used in both of the analyses, separately for the pre- and post-treatment periods. Observations are defined at the household–week–period level (for example, average weekly electricity consumption during the morning off-peak period). For ease of interpretation, these outcomes are scaled to annualized kWh equivalents, although the underlying variation exploited in the estimation is at the weekly level. The table shows that households that adopt the heat pump tariff consume, on average, more electricity than households in the heat pump installation sample, consistent with adopters tending to live in larger dwellings with higher baseline energy demand (see Table A.19 for a comparison of our samples). For the tariff adoption sample, average electricity consumption is higher in the post-adoption periods; however, these level differences are not reflected in the Callaway and Sant'Anna

(2021) estimates reported below, which exploit within-household changes over time. This pattern is consistent with the possibility that households adopt other low-carbon technologies or undertake energy-related investments during the sample period. We conduct a range of robustness checks and balance tests to ensure that such compositional changes do not bias the estimated treatment effects (see Section A.5.1).

**Table 1:** Summary statistics before and after adoption

| | Period | Obs | Mean | SD |
|---|---|---|---|---|
| **Panel A: Heat Pump Adoption Sample** | | | | |
| Electricity consumption | Pre | 77,348 | 5,049.45 | 4,148.30 |
| | Post | 25,033 | 8,612.40 | 5,547.98 |
| Gas consumption | Pre | 64,429 | 10,097.86 | 9,922.03 |
| | Post | 21,668 | 483.50 | 2,415.04 |
| **Panel B: Tariff Adoption Sample** | | | | |
| Overall (all 24 hours) | Pre | 235,857 | 6,629.36 | 6,381.11 |
| | Post | 200,505 | 8,602.01 | 6,799.14 |
| Morning off-peak (04:00–07:00) | Pre | 235,703 | 795.45 | 987.18 |
| | Post | 200,491 | 2,258.98 | 2,244.23 |
| Afternoon off-peak (13:00–16:00) | Pre | 235,716 | 696.64 | 802.26 |
| | Post | 200,493 | 1,549.87 | 1,644.41 |
| Peak (16:00–19:00) | Pre | 235,712 | 960.85 | 1,019.52 |
| | Post | 200,493 | 684.68 | 836.04 |
| Standard rate, all other hours | Pre | 235,857 | 4,193.73 | 4,059.93 |
| | Post | 200,505 | 4,107.67 | 4,039.33 |

**Note:** This table reports summary statistics for the two estimation samples. All figures are annual consumption in kWh, obtained by annualizing the weekly consumption (panel A) and half-hourly (panel B) used in the difference-in-differences models. "Pre" is the period before adoption and "Post" the period after; weeks within the anticipation window are excluded. *Obs* is the number of household–period observations, and Mean and SD are computed across those observations. *Panel A* is the heat pump adoption sample and reports household electricity and gas use. *Panel B* is the heat-pump time-of-use tariff adoption sample: two cheap off-peak windows in the morning (04:00–07:00) and the afternoon (13:00–16:00), a peak window (16:00–19:00) priced above the standard rate, and the standard rate that applies in all other hours (07:00–13:00 and 19:00–04:00); "Overall" covers all 24 hours.

## 2.1 Impact of Heat Pump Installation on Energy Consumption

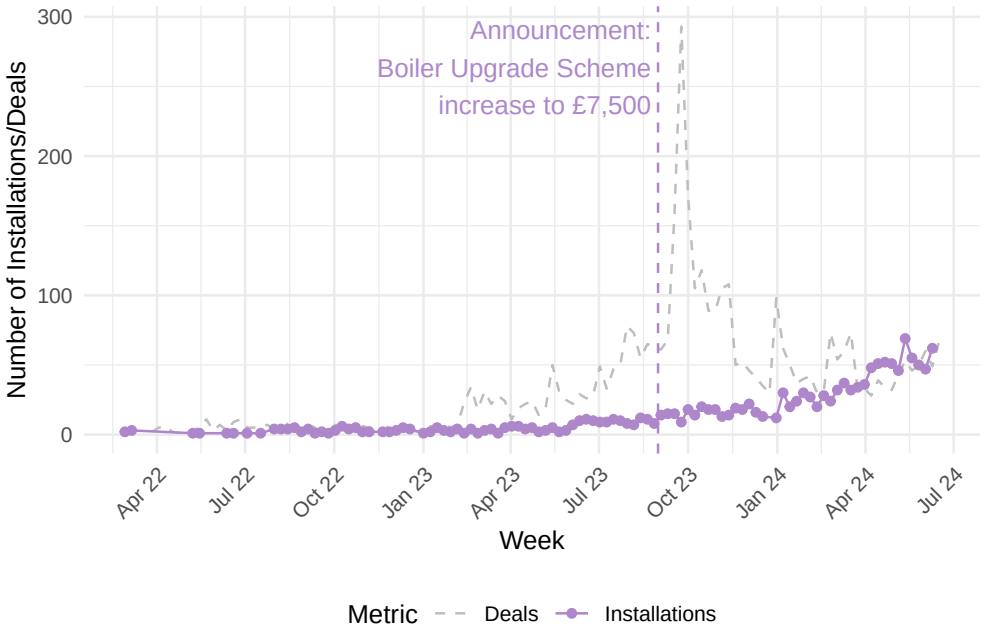
### 2.1.1 Set-up and Identification Strategy

This section outlines our approach to identifying the effects of heat pump installation. Simply comparing adopters to non-adopters risks bias, since households that adopt may differ in unobservable ways correlated with their potential outcomes. We instead exploit the staggered adoption of heat pumps across Great Britain among households that purchased directly from a large energy supplier, using households that adopt later as a counterfactual control group. While it is unusual for a utility to sell and install heat pumps, this one has pursued vertical integration, providing installations, maintenance, and tariff provision as part of its broader commercial and decarbonization strategy. It began taking orders in December 2021 and installing in February 2022. Figure 1 shows the rollout over time, where a deal is a household paying a refundable deposit to obtain a heat pump.

A delay between households making a deal for the heat pump and installation, due to installa-

tion constraints, provides us with a natural experiment to understand the impact of heat pumps on energy demand for those who want a heat pump. We see evidence that deals increased markedly in September 2023, when the Government announced its plan to increase the level of the main subsidy for heat pumps (the Boiler Upgrade Scheme), from £5,000 to £7,500 per air-source heat pump. However, because of the delay between deal and installation, we see a slower and steadier increase in heat pump *installations* both before and after September 2023.

**Figure 1:** Weekly Deals and Installations




**Note:** The solid purple line and dots show weekly heat pump adoptions, comprising our sample of 1,308 adopters. The dashed gray line shows *deals*, i.e., reservations made by paying a refundable deposit, over the same period. The heat pump subsidy increased from £5,000 to £7,500 in October 2023, following an announcement in September 2023.

We use the universe of households that adopted a heat pump with the partner utility as their installer. There are 1,308 such households who adopted a heat pump as early as 28 February 2022 or as late as 15 June 2024, and who have sufficient consumption data for analysis. Households who have a deal but no installation as of 15 June 2024 are akin to never treated in our sample; we exclude these 634 households from our main analysis but include them in a robustness check we present in Table A.2. We examine consumption from 13 December 2021 through 15 June 2024.

Almost all households in our sample took advantage of the heat pump subsidy. On average, it was worth £6,800 to the households in our sample. This is a weighted average of the £5,000 subsidy level and the later, more generous, £7,500 level. The average cost of an air-source heat pump installation since the subsidy increased to £7,500 in October 2023 is £12,713, based on MCS's monthly average costs from November 2023 to June 2024 (Microgeneration Certification Scheme (MCS) 2024). Because MCS certification is required for subsidy eligibility, these figures cover all eligible installations, implying an average private cost of £5,213 after the subsidy. Heat pumps in

our sample have an average size of 8 kW, which is typical of domestic heat pumps in the UK.

**Callaway and Sant'Anna (2021) in This Context:** Staggered difference-in-differences models may produce biased estimates if treatment effects are heterogeneous across units and over time (Callaway and Sant'Anna 2021; Goodman-Bacon 2021). This might happen in our setup because: (a) households may get better at using their heat pump over time; and (b) households could also adopt more low-carbon technologies to complement their heat pump usage. With this in mind, we calculate Callaway and Sant'Anna (2021) (CS) difference-in-differences estimators. We do this by aggregating the data at weekly level. Our cohorts are divided by their week of heat pump adoption, and the unit of observation is the average consumption for household $i$ in the week of installation (cohort) $g$ in week $t$.

In settings with staggered adoption and time-varying treatment effects, conventional summaries, such as simple average treatment effects or event-time dynamic average treatment effects on the treated (ATTs), can be difficult to interpret. These summaries often conflate true treatment heterogeneity with variation in cohort composition and exposure length. In our setting, where treatment effects are strongly seasonal, this issue is particularly acute: for example, the simple ATT depends mechanically on how much of the post-adoption period households spend in colder versus milder months, rather than capturing a stable annual impact of heat pump adoption. As a result, commonly reported estimates can vary substantially across samples or horizons even when underlying behavioral responses are unchanged.

To address this problem, we apply what we believe is a new and practically important aggregation approach that has not previously been used to summarize treatment effects in staggered adoption settings with pronounced seasonality. Specifically, we exploit calendar-time ATTs as a constructive aggregation device that preserves the seasonal structure of treatment effects. By estimating impacts at each point in calendar time and then aggregating over a full annual cycle, we recover yearly effects on electricity and gas consumption that are interpretable and comparatively stable, up to natural variation in winter severity. This approach separates seasonality in treatment effects from differences in exposure timing across cohorts, yielding annual estimates that more closely reflect the policy-relevant impact of heat pump adoption. Our main results use the most recent 12 months of data, which provide the greatest precision in our setting. As a robustness check, we also report 12-month rolling averages constructed from calendar-time ATTs in Figure A.1.

We believe this approach offers a practical solution for many empirical settings characterized by staggered adoption and pronounced seasonal responses, common in energy, environmental, and climate policies, where the object of interest is an annual or long-run impact rather than an exposure-weighted average over event time. In such contexts, calendar-time aggregation provides a transparent way to summarize heterogeneous treatment effects without obscuring their underlying seasonal structure.

The validity of our identification strategy rests on four assumptions. First, we have irreversible

treatment, so once a household installs a heat pump, it does not revert to its previous heating system, which is a highly plausible assumption given that subsidy receipt requires the removal of any existing gas boiler. Second, we have random timing of treatment driven by installation constraints, so selection into receiving a heat pump earlier versus later is not correlated with underlying trends in energy use; the average waiting time in our sample is 28 weeks, creating a natural experiment in which households awaiting installation, who nonetheless *want* to adopt, form a credible counterfactual. Third, we have no anticipation, so households do not change their behavior in anticipation of heat pump installation, or if they do, this anticipation is limited to a clearly defined period; we use an anticipation period of four weeks.[1] Fourth, we assume unconditional parallel trends based on "not-yet-treated" groups, ensuring that the comparison between treated and untreated groups is valid, as the trends in outcomes would have been parallel in the absence of treatment.

### 2.1.2 Dynamic ATTs

We begin by examining the dynamic effects of heat pump adoption on electricity and gas consumption using an event-study framework. We show dynamic ATTs of heat pump installation on electricity and gas consumption in Figure 2. We find that the impact of heat pump adoption is effectively immediate, with a sharp reduction in gas consumption and a corresponding increase in electricity use at the time of installation. There is little evidence of differential trends prior to adoption. We use a universal base period with a four-week anticipation period; as discussed in Section 2.1.1, we see evidence of an anticipation period of up to one month before adoption (Figure A.4).

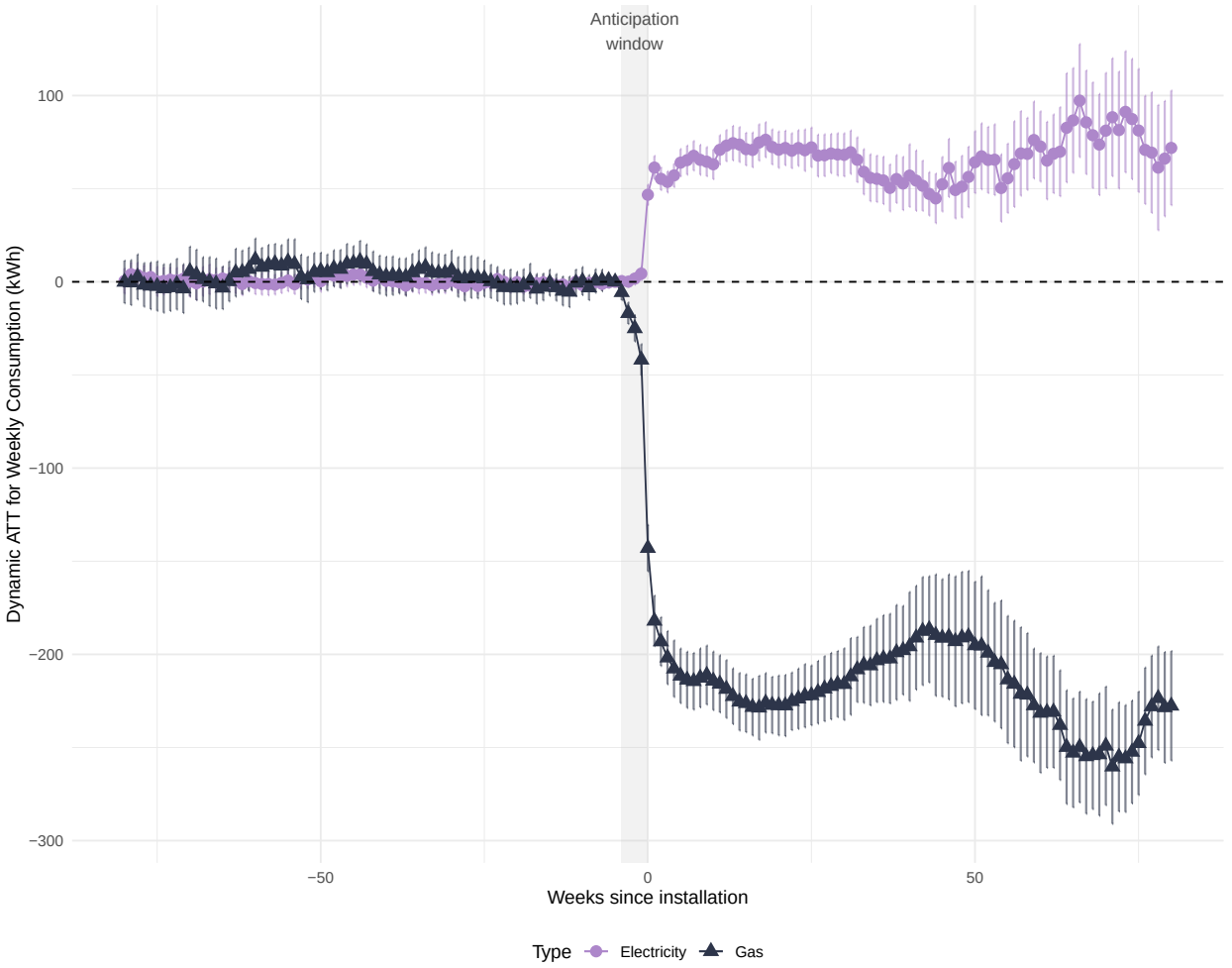
The dynamic ATTs exhibit flat pre-trends, reducing concerns about pre-treatment divergence and increasing our confidence in our overall identification strategy. Notably, our large sample gives high power to detect modest pre-trend differences, mitigating low-power critiques (e.g., Freyaldenhoven et al. 2019; Roth 2022).

In a staggered-adoption setting with heterogeneous effects, event-time patterns can reflect cohort composition and exposure timing as much as true post-adoption dynamics. In our setting, seasonality is a key driver of this issue: event-time estimates may embed seasonal variation because cohorts adopt at different times of the year. Since the largest cohorts tend to adopt at the beginning of winter (see Figure A.7), the early post-adoption weeks predominantly reflect outcomes from these groups, embedding a systematic seasonal pattern into the event study results even though later cohorts may adopt in different seasons.

A related concern is that installation timing may not be random across households. In practice, installations could occur earlier for reasons such as equipment failure, property characteristics, or

[1] We see evidence of an anticipation period up to a month, as can be seen in Figure A.4. Specifically, heat pump impacts increase slightly when we use longer anticipation periods. This is especially the case for the extent to which heat pumps cause a reduction in gas consumption. We hypothesize that this pattern may be due to people removing their gas boilers more than one week before heat pump installation, even up to a month before heat pump installation. Households might then use portable electric heaters as temporary substitutes during this period.

**Figure 2:** Gas and Electricity Dynamic ATTs




**Note:** This figure shows how the estimated effect of heat pump adoption on energy consumption evolves over time. Electricity (purple) and gas (dark gray) effects are shown separately. The y-axis shows the weekly consumption effect in kWh; the x-axis shows weeks relative to installation, with week 0 the first full week after installation. The gray box marks a four-week anticipation window running from four weeks before installation through the installation week. Estimates come from a Callaway and Sant'Anna (2021) event-study design comparing treated and not-yet-treated households; no additional controls are included.

household readiness, which may themselves correlate with energy use patterns. Our difference-in-differences design partly mitigates this issue by comparing households relative to their own pre-treatment outcomes and to contemporaneous controls that also adopt during the sample period but at different times. The flat pre-trends in Figure 2 provide further reassurance, although unobserved factors correlated with installation timing cannot be ruled out entirely.

It is useful to clarify why the event-study pre-trends exhibit little seasonality. Under the not-yet-treated estimator, comparisons are always made within the same calendar week between units that are treated in that week and units that have not yet been treated. In the pre-adoption periods for a given cohort, both the "earlier" and "later" adopters are untreated in the relevant weeks, so any seasonal patterns in the outcome, such as fluctuations in energy use, are common across

groups and difference out mechanically. The control principle is the same in every calendar week; what changes over time is whether a given cohort has begun treatment. After adoption, the same comparison rule applies, but treated units are now exposed to the policy while the not-yet-treated units remain untreated. If the treatment effect itself varies over calendar time, for example because the policy interacts with seasonal demand conditions, then the estimated post-adoption average treatment effects can display seasonal patterns. This reflects seasonally heterogeneous treatment effects and the weighting of different cohorts and exposure lengths, rather than differential seasonality in untreated outcomes or a violation of parallel trends.

#### 2.1.3 Seasonal Variation

To directly capture the seasonal nature of heat pump impacts, we next examine treatment effects in calendar time. Calendar-time ATTs measure the effect of heat pump adoption at each point in the year, pooling across all cohorts that are treated in that period. This perspective is particularly informative in our setting, where space-heating demand, and thus both electricity increases and gas reductions, varies sharply with outdoor conditions.

We show calendar ATTs in Figure 3. During winter, the calendar ATT increase in electricity consumption reaches over 200 kWh per week. Meanwhile, it drops to almost zero between April and October. The decrease in gas consumption is larger in magnitude, but shows similar seasonal patterns, with very little decrease during warm months and a large decrease during the winter.

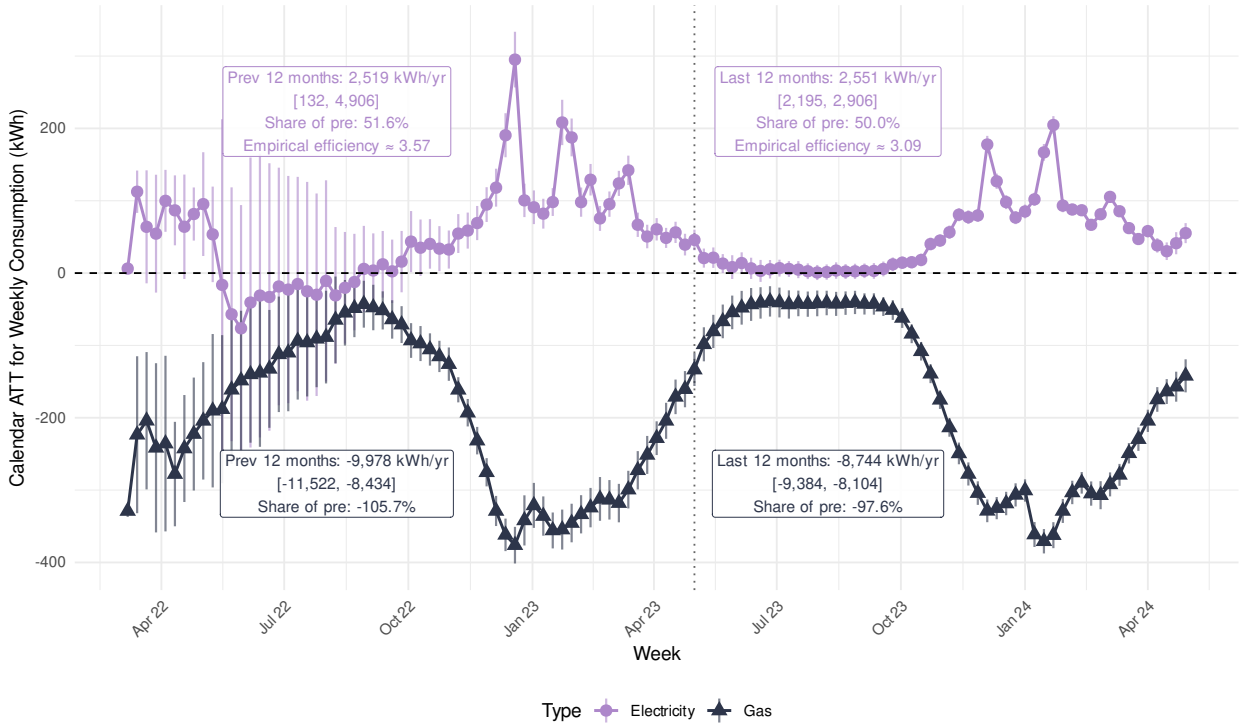
#### 2.1.4 Main Results

To construct annual impacts of heat pump adoption on electricity and gas consumption, we aggregate calendar-time ATTs (shown in Section 2.1.3). As discussed in Section 2.1.1, we sum weekly calendar ATTs over a 12-month window, yielding annualized changes in energy use that preserve the seasonal structure of treatment effects while avoiding distortions arising from cohort composition and time spent in treatment. We focus on the most recent 12 months of data, which provide the most precise estimates in our setting.

We find that heat pump adoption increases electricity consumption by 2,551 kWh per year and reduces gas consumption by 8,744 kWh per year. To aid interpretation, we also examine these effects relative to pre-installation energy use, measured as a 12-month rolling average of electricity and gas consumption among not-yet-treated households observed in the same calendar weeks. Relative to this baseline, the electricity increase corresponds to 50% of pre-installation electricity consumption (5,099 kWh), while the reduction in gas consumption corresponds to 98% of pre-installation gas use (8,963 kWh).

The results remain consistent across different choices of rolling average windows, indicating the robustness of the annualized impact estimates. For example, aggregating the preceding 12 months

**Figure 3:** Gas and Electricity Calendar ATTs




**Note:** This figure shows how the estimated effect of heat pump adoption on energy consumption varies across weeks of the year. Electricity (purple) and gas (dark gray) effects are shown separately. The y-axis shows the weekly consumption effect in kWh; the x-axis shows the week of the year.

yields similar magnitudes (electricity: 2,519 kWh/yr; gas: -9,978 kWh/yr), with variation mainly driven by winter severity. Twelve-month rolling averages of these annual impacts are shown in Figure A.1 and demonstrate smooth, stable annual estimates over time.

**Simple ATTs and Estimator Comparisons:** To complement the annualized estimates from rolling average calendar-ATTs, we also report simple average treatment effects from the CS estimator and their two-way fixed-effects (TWFE) equivalents in Table A.1. These estimates summarize impacts on weekly electricity consumption for all households in our sample, and on weekly gas consumption for the subset of households with natural gas service. For ease of interpretation, coefficients and standard errors are scaled to annual units by multiplying weekly estimates by 52.

As noted above, these simple ATTs average treatment effects across cohorts and post-adoption periods and therefore mix observations from different seasons and different lengths of exposure. As a result, they are not our preferred estimates of annual impacts in a setting with strongly seasonal and heterogeneous treatment effects. Instead, we use them as a diagnostic benchmark and as a point of comparison with alternative estimators and samples.

These estimates nonetheless are similar to our preferred specification. Heat pump adoption is associated with higher electricity consumption and a large reduction in gas use, consistent with

the substitution patterns documented using calendar-time ATTs. The magnitude of these simple averages is outside the range implied by our annual calendar-based estimates, but similar once differences in weighting and exposure are taken into account.

**Gas Consumption and Sample Coverage:** Most households in our sample replace a gas boiler when installing a heat pump, as confirmed by administrative survey data collected at the time of installation. We therefore examine gas consumption for households that had an active gas account prior to installation and at least four weeks of observed gas use in the pre-treatment period.

Gas consumption data are unavailable for households supplied by other retailers or without a gas connection, and gas smart meter readings are more frequently missing than electricity readings. As a result, the gas analysis relies on a smaller sample. Reassuringly, when we restrict the electricity analysis to this subsample, estimated electricity impacts remain very similar to those in the full sample, indicating that sample selection is unlikely to drive the results (see Table A.3 and Figure A.3).

**Relative Impacts Across Fuels and Emissions:** The reduction in gas consumption is over three times larger than the increase in electricity consumption in absolute kWh terms. Final energy use does not translate one-to-one into primary energy because electricity generation involves conversion losses, whereas gas boilers convert fuel to heat at higher efficiencies. However, the marginal carbon intensities of electricity and natural gas in Great Britain are currently of a similar order of magnitude (Department for Energy Security and Net Zero 2023*b*; UK Government 2024). As a result, the observed substitution from gas to electricity translates into an immediate and substantial reduction in carbon emissions from residential heating.

We acknowledge that some forms of backup heating may be imperfectly observed in our data. Electric resistive backup heating would be mechanically included in measured electricity consumption, biasing estimated electricity increases upward and implied energy efficiency downward, while non-electric backup heating (e.g., gas or solid fuels) would reduce observed electricity demand, biasing estimated electricity increases downward and implied energy efficiency upward. If reliance on backup heating rises on very cold days, these channels could affect implied energy efficiency estimates at the cold end of the temperature distribution.

**Operating Costs:** Because electricity prices per kWh substantially exceed gas prices in the UK, heat pump operating costs are close to parity with those of the gas boilers they replace. Using contemporaneous retail prices[2], we estimate a modest saving of around £10 per year relative to gas boiler heating; this turns into a cost increase of around £97 per year if the household remains

[2] As of 15 June 2024, the average electricity price per kWh was £0.2450 versus a gas price per kWh of £0.0604 using the utility's standard tariff (Octopus Energy 2024). UK Government (2024) retail price estimates show similar differences in fuel prices, although electricity and gas price per kWh are both higher than the utility's own prices.

connected to the gas grid, due to a daily gas standing charge. These figures should be interpreted as indicative, as operating costs depend on tariff choice. In particular, heat pump–specific time-of-use tariffs can substantially reduce electricity costs, an issue we explore in more detail in the heat pump tariff analysis (Section 2.2). In addition, we discuss the impact of policy and regulatory choices on these operating costs in Section 3.2.

#### 2.1.5 Temperature Interactions

To understand the mechanisms behind the seasonal patterns documented above, we next examine how treatment effects vary with outdoor temperature. For this and other heterogeneity analyses, we use TWFE models, since the CS estimator does not readily accommodate finely binned treatment–temperature interactions. As in our CS analyses, we estimate separate models for weekly electricity and gas consumption.

In these models, we interact heat pump installation status with weekly average outdoor temperature, measured at the Grid Supply Point (GSP) Group level and rounded to the nearest degree Celsius. Temperature is treated as a categorical variable and bounded between 0°C and 25°C. GSP Groups are geographic zones commonly used in the Great Britain energy sector, and we assign each household the temperature recorded at the highest-quality weather station in its group. All specifications include household and week fixed effects, and standard errors are clustered at the household level.

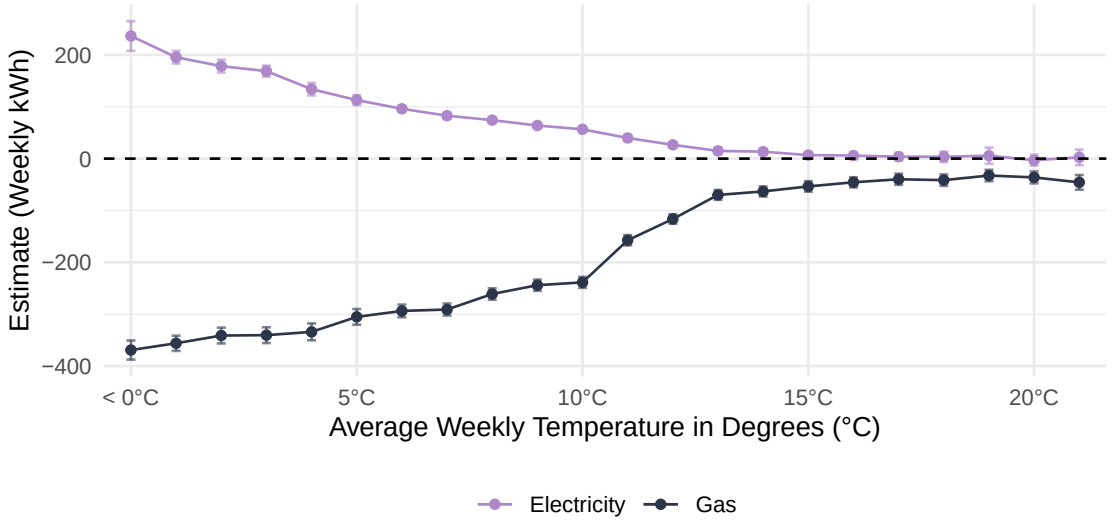
By interacting heat pump installation status with temperature, while also controlling for temperature on its own, we isolate how the impact of heat pump adoption varies across outdoor conditions. In this specification, treatment effects are identified through these interaction terms rather than through a single average installation effect. The resulting TWFE estimates characterize the shape of the temperature-response relationship but not its absolute magnitude, since staggered adoption can introduce modest bias (around 7% for electricity and gas; see Table A.1). For this reason, our preferred treatment effects remain our CS estimates.

We plot these temperature-varying effects in Figure 4. Electricity impacts rise sharply as temperature falls below 15°C, and are close to zero in warmer weeks, consistent with air-to-water heat pumps operating on the space-heating margin. Gas reductions display a similarly winter-peaked pattern. This is natural in the UK context, where air-to-water heat pumps typically replace gas boilers in "wet" central heating systems and do not provide active cooling: electricity use increases over the heating-relevant range, with no corresponding summer cooling effect. The persistent reduction in gas use at higher temperatures likely reflects broader technological and behavioral changes associated with adoption, such as electrification of domestic hot water via a hot water cylinder and reduced reliance on gas appliances, rather than space heating *per se*.

These temperature profiles also clarify the external validity of our results. They speak most directly to settings where heat pump adoption entails a near-complete switch from fossil fuel boil-

ers to air-to-water systems in a heating-dominated climate with limited cooling demand (e.g., the UK and other countries with hydronic radiator systems). In settings dominated by air-to-air heat pumps that jointly provide heating and cooling, often alongside continued fossil heating, one would expect weaker winter substitution away from gas or oil and stronger electricity increases on hot days due to cooling. Our findings should therefore be interpreted as evidence on whole-system electrification of heating in cold and temperate climates, rather than on the joint heating-and-cooling margin relevant in warmer regions.

**Figure 4:** Heat Pump Impacts on Gas and Electricity Demand, by Outside Temperature




**Note:** This figure shows how the estimated effect of heat pump adoption on energy consumption varies with outside temperature. Estimates come from a TWFE model interacting heat pump installation status with average weekly outside temperature in °C (binned into 0–25°C, with tails grouped). The dependent variable is electricity or gas consumption in kWh. We control separately for temperature and include household and time fixed effects, so interaction terms capture the additional impact of heat pump installation at each temperature level. Standard errors are clustered at the household level. The sample includes 1,308 households for the electricity analysis and 1,110 for the gas analysis.

We show in Figure A.8 that this relationship holds true across different times of day using daily electricity data (note that the temporal nature of this analysis is slightly finer grained than the main temperature interactions, which are based on weekly averages), including the peak time (which is between 4pm and 7pm in the UK).[3] Lower outside temperatures drive higher electricity usage due to increased heating demand.

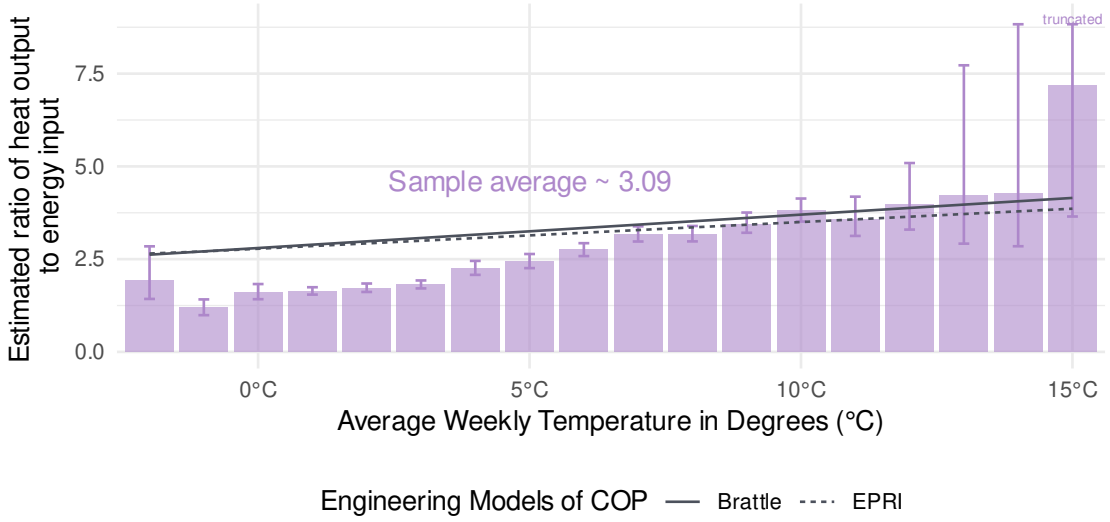
### 2.1.6 Empirical Heat Pump Energy Efficiency

We can also calculate an empirical estimate of heat pump efficiency by comparing how adoption changes annual gas and electricity consumption. In engineering terms, the Coefficient of Perfor-

[3]For convenience, we show these impacts by the different rate periods of the heat pump tariff investigated in Section 2.2.2, but it should be noted that the majority of our heat pump adopting households are on tariffs other than the heat pump time-of-use tariff analyzed later.

mance (COP) is the ratio of useful heat output to electricity input. Because we do not observe heat output directly, we cannot estimate a physical COP. Instead, we infer an empirical energy efficiency measure from observed fuel substitution: we take the absolute ratio of the annual gas reduction to the annual electricity increase implied by our calendar-time ATT aggregation, and multiply by 0.9 to adjust for the fact that gas boilers convert fuel to useful heat with roughly 90% energy efficiency (typically 78–97%; see Department of Energy and Climate Change 2009; U.S. Department of Energy 2023).

**Figure 5:** Estimated Gas (kWh Reduction) to Electricity (kWh Increase) Energy Demand Ratio




**Note:** This figure compares the empirical energy efficiency of heat pumps at different temperatures with engineering predictions. Temperature interactions are estimated using TWFE because the CS estimator does not support treatment–temperature interaction terms. Headline annual impacts and the annual implied empirical heat pump energy efficiency are constructed separately from calendar-time ATTs. The figure is restricted to temperatures at or below 15°C to avoid unstable ratios when interaction coefficients approach zero. We do not observe households with weekly average temperature below -2°C in our dataset. Error bars denote 95% clustered bootstrap confidence intervals.

This empirical energy efficiency captures two components: (i) the relative energy efficiency of heat pumps versus gas boilers in delivering heat, and (ii) any behavioral responses to adoption (e.g., changes in heat demand) that affect observed energy use. Using the CS-based annual impacts in Figure 3, we obtain an implied empirical heat pump energy efficiency of 3.09. This value is consistent with expected performance of air-to-water heat pumps in the UK, and falls between seasonal performance factors (SPFs) reported for participants in the UK Government's Electrification of Heat trial (average SPF 2.81) and the average SPF of households contributing data to Heatpump-Monitor.org (average SPF 3.86; see Rosenow et al. 2026 for discussion of why these two averages differ). Note that the UK Climate Change Committee uses the former energy efficiency, 2.81, as its main assumption for heat pump energy efficiency in its Seventh Carbon Budget modeling (Climate Change Committee 2025, p. 170).

As discussed previously, our main results are based on a 12-month rolling average of calendar

ATTs, and thus our empirical energy efficiency estimate is as well. As shown in Figure A.1, 12-month rolling averages of the annual impacts, and the implied empirical energy efficiency, remain consistent over time.

We can also use the temperature-interaction estimates from Section 2.1.5 to construct a *temperature-varying* empirical energy efficiency, shown in Figure 5. We focus on temperatures at or below 15°C, where coefficients are stable and the ratio is well-behaved. The empirical energy efficiency is lowest in very cold weather and rises as temperatures become milder, remaining above 1.15 even at 0°C. The lowest observed mean weekly temperature is -2°C.

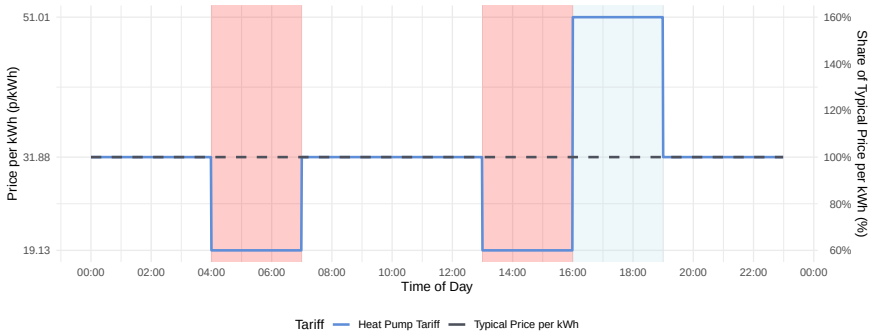
Overall, our empirical energy efficiency estimates are broadly consistent with engineering projections in Electric Power Research Institute (2018) and Sergici et al. (2023). These estimates, and the empirical pattern we document, are consistent with the basic thermodynamics of heat pumps: when outdoor temperatures are lower, the system must move heat across a larger temperature gap, which mechanically reduces energy efficiency. That said, we find somewhat lower energy efficiency in very cold weather and higher energy efficiency at milder temperatures. A likely explanation lies in how installed heat pump systems are actually operated, rather than in the basic thermodynamics described above. Most systems use a *weather compensation* curve: as outdoor temperature falls, the controller automatically raises the flow (water) temperature delivered to radiators to maintain the target indoor temperature, rather than relying solely on a room thermostat to react after the fact. Because a higher flow temperature itself reduces the coefficient of performance, independent of the indoor-outdoor temperature gap, weather compensation causes efficiency to decline faster in cold weather than a model based on temperature gap alone would predict. Engineering coefficient-of-performance estimates are typically reported at a fixed flow temperature and therefore do not capture this effect. Households may compound this further by raising thermostat setpoints during especially cold spells to maintain comfort. Our findings also partially align with Watson et al. (2023), who directly measure heat output from the heat pump divided by all electricity input to the heat pump to compute COP using the conventional definition. While their approach isolates device-level performance, our empirical measure additionally reflects behavioral adjustment following adoption. Similar to our results, they document a non-linear relationship between outdoor temperature and energy efficiency; however, whereas they find lower energy efficiency at higher temperatures, we observe higher implied energy efficiency in milder conditions.

## 2.2 Impact of Heat Pump Tariff on Electricity Consumption

Having established the impacts of heat pump installation on electricity and gas demand, we now turn to our second main analysis: the effect of heat pump–specific electricity pricing on household electricity consumption. We study *Cosy Octopus*, a time-of-use tariff introduced in 2022 that targets heat pump households and is designed to incentivize load shifting away from peak periods, when electricity generation is most carbon-intensive. This section evaluates whether targeted pricing can

meaningfully reshape consumption patterns among electrified-heating households, complementing the technology adoption effects documented above.

**Figure 6:** Price per kWh by Rate Period




**Note:** We show the price per kWh for heat pump tariff adopting households by time of day (blue line) compared to the typical price per kWh of the utility's standard tariff in North West England. The tariff was first made available to households on 13 December 2022. Unless otherwise noted, price levels in the paper use the North West England tariff schedule as of 15 June 2024. The two off-peak 40%-reduction periods relative to the typical price per kWh are shaded light red, while the peak period with 160% of the typical price per kWh is shaded light blue. Prices per kWh vary slightly by region; there are 14 tariff-pricing regions in Great Britain. In all 14 regions, the overall structure of the heat pump tariff is similar: 40% lower than the typical price per kWh for the off-peak periods, 60% higher than the typical price per kWh for the peak period, and the typical price per kWh for all other hours of the day. Prices can also change over time; but, again, the overall structure of the tariff remains as in Figure 6 during the period of our analysis, except in the final two months of our analysis period when the peak is 151% of the standard price per kWh and off-peak is 45% lower. (See Section A.4.2 for further detail.)

The price per kWh faced by adopting households differs by hour of the day, as we show in Figure 6. Specifically, the heat pump tariff features two off-peak periods and one peak period. All remaining hours are charged at the utility's standard price per kWh. For most of the analysis period, off-peak prices are set at 60% of the standard tariff, while the peak period is priced at 160% of the standard rate. We show average wholesale prices across the two off-peak and peak periods in 2022, 2023, and the first half of 2024 in Table A.8 based on data from LCP Delta. The first off-peak period is in the morning, from 4am to 7am, and the second in the early afternoon, from 1pm to 4pm. The peak period is from 4pm to 7pm. The tariff's prices within a day are designed to broadly correlate with wholesale market prices. The tariff structure is also designed to align with how households use heat pumps. In particular, the two off-peak periods allow households to pre-heat their homes ahead of the morning and evening periods of higher prices, while the peak period imposes additional costs on electricity use during the evening.

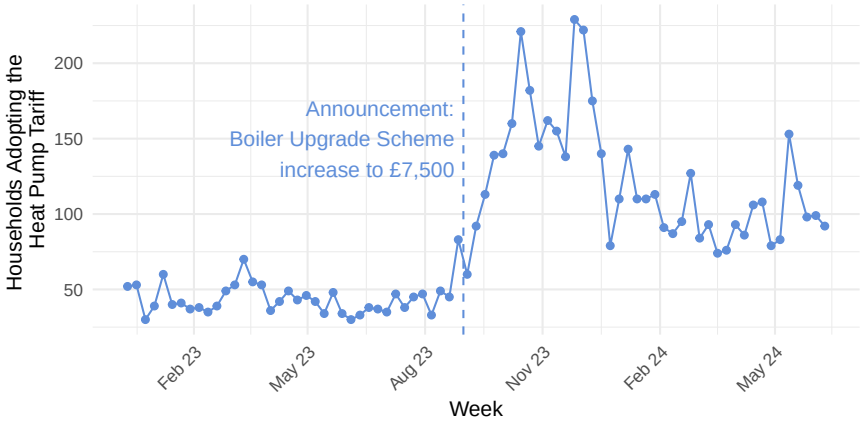
We use the full universe of households who switched to the heat pump tariff: 6,631 households who adopted the tariff between 13 December 2022 and 15 June 2024. Most households in this sample are not in the heat pump adoption sample. Many households had their heat pumps installed before 2022 or used a third-party installer, and therefore did not obtain their heat pump through the utility.

Our analysis covers the same time period as the heat pump adoption analysis, focusing on

consumption data from 13 December 2021 through to 15 June 2024. Our analysis measures households' consumption for seven months after adoption, on average, not just the weeks immediately before and after adoption. Many households in our sample have two winters of data on the tariff.

### 2.2.1 Timeline of Heat Pump Tariff Adoption

**Figure 7:** Weekly Adoption of the Heat Pump Tariff




**Note:** We show weekly switches among the 6,631 households in our heat pump tariff analysis sample.

As with our heat pump analysis, we employ CS models to assess the impact of the tariff on consumption during different periods of the day. Again, our identification strategy relies on the parallel trends assumption: in particular, that the electricity consumption of households that adopted earlier would have followed a parallel trend to control households (in this case, those who adopted later) in the absence of adoption of the tariff. We also assume that once households adopt the tariff, they do not leave. This assumption is not as strong in the case of tariff adoption as for heat pump installation; we conduct robustness checks on leavers in Table A.13.

### 2.2.2 Main Results

Switching to the heat pump tariff induces large and persistent shifts in electricity use across the day. To summarize these effects in a way that is directly comparable to the heat pump adoption results, we report 12-month average calendar-time treatment effects, computed over the final year of available data (June 2023–June 2024). As shown in Table 2, households substantially increase electricity consumption during incentivized off-peak periods and reduce demand during peak and standard-rate hours, while leaving total daily electricity use essentially unchanged.

On average over this 12-month window, households increase electricity consumption by 0.43 kWh per half-hour during the early-morning off-peak period (4am–7am), corresponding to a 123% increase relative to the mean consumption level of not-yet-treated households in these hours. Elec-

tricity use also rises during the afternoon off-peak window (1pm–4pm), by 0.26 kWh per half-hour on average, equivalent to a 79% increase relative to the non-treated baseline. In contrast, electricity demand during the peak period (4pm–7pm) falls by 0.21 kWh per half-hour, around 47% of baseline consumption, while usage during standard-rate hours declines by 0.10 kWh per half-hour, a reduction of approximately 28%. Averaged across the day, the change in overall electricity consumption is statistically indistinguishable from zero, suggesting that the tariff primarily induces load shifting rather than changes in overall energy demand. For comparison, we report corresponding simple ATTs and TWFE estimates in Table A.9, which yield similar qualitative patterns.

The tariff-induced reallocation of electricity demand is highly persistent, rather than a short-lived behavioral response. Aside from the seasonal variation discussed in Section 2.2.3, we observe little attenuation in the magnitude of these effects over time (see Figure A.20). This durability is consistent with a substantial role for automated or easily scheduled responses to prices: tariff survey evidence in Section A.4.5 indicates that 64% of households that respond to the tariff do so through some form of automation. That said, we cannot rule out that tariff adoption coincides with increased attention to electricity use, marketing, or other salience effects at the time of switching. Our estimates should therefore be interpreted as reduced-form treatment effects that reflect the combined influence of price incentives, automation, and any contemporaneous shifts in household attention.

Consistent with this interpretation, heterogeneity analysis in Section A.4.6 shows larger absolute responses among households that report owning flexible low-carbon technologies, such as electric vehicles or home batteries, which can more easily automate load shifting. Among households without these technologies, the absolute magnitude of the tariff response is roughly half as large (comparing the `Contract Active = 1` coefficients in Tables A.9 and A.11). Because ownership of such technologies is correlated with other household characteristics (e.g., home size or baseline electricity demand), these differences should be interpreted as descriptive rather than causal evidence on mechanisms.

The magnitude of this load-shifting response is large relative to much of the existing time-of-use and dynamic pricing literature (e.g., Borenstein 2007; Ito et al. 2023; Wolak 2011). That said, some studies show similar large demand response, especially when pairing incentives with technology; see Bailey et al. (2025); Blonz et al. (2025); Bollinger and Hartmann (2020); Gillan (2017); Jessoe and Rapson (2014). We acknowledge that our results may reflect selection into treatment by structural winners, and even more so by low-shifting-cost households: those who adopted the tariff so far have larger homes and may have more technologically engaged members of the household, well placed to automate load shifting. As tariff adoption extends beyond these earlier adopters to the early majority, including smaller homes and those with less familiarity with automation, we expect the absolute magnitude of load shifting to fall, though the response should remain meaningful once scaled to heat pump and home size.

The within-day shifts imply substantial changes in electricity consumption across hours in re-

**Table 2:** Heat Pump Tariff Adoption on Half Hourly Electricity Consumption in kWh

| Rate period | Mean ATT (se) | NYT baseline | Share (ATT/NYT) |
|---|---|---|---|
| Morning Off-peak | 0.434 (0.013) | 0.353 | 123.0% |
| Afternoon Off-peak | 0.258 (0.009) | 0.329 | 78.6% |
| Peak Rate | -0.207 (0.010) | 0.443 | -46.8% |
| Other | -0.103 (0.007) | 0.371 | -27.9% |
| Overall | 0.001 (0.007) | 0.371 | 0.3% |

Calendar window: June 2023 to June 2024.

**Note:** This table summarizes calendar-time average treatment effects on the treated (ATT) for the heat pump tariff adoption analysis over June 2023 to June 2024. Weekly calendar ATTs are obtained from the Callaway–Sant'Anna estimator aggregated to calendar time. For each rate period, we report the *mean* calendar ATT over this period ("Mean ATT"). The standard error in parentheses is computed as $\sqrt{w'\Sigma w}$, where $\Sigma$ is the covariance matrix of weekly calendar ATT estimates (from the did influence-function/cluster bootstrap covariance) and $w_t = 1/T$ over the selected window. "NYT baseline" is the mean outcome among not-yet-treated households over the same weeks. "Share (ATT/NYT)" reports the ratio of the mean ATT to the NYT baseline, expressed as a percentage. Outcomes are measured in kWh per half-hour.

sponse to the tariff's relative prices. However, these responses should not be interpreted as extremely large price elasticities in the conventional sense, which measure the response of total electricity use to a single marginal price (Ito 2014). Instead, our setting is more relevant to measures of interhour elasticity of substitution (Imelda et al. 2024): households reallocate electricity use across hours in response to a vector of simultaneously changing prices, while total daily consumption remains essentially unchanged.

This distinction matters because a back-of-the-envelope elasticity computed as $\varepsilon \approx (\Delta q/q)/(\Delta p/p)$ within a given period, such as an implied elasticity of about $-3.1$ in the morning off-peak period and $-0.78$ in the peak period, does not isolate a structural own-price elasticity. Rather, it reflects a reduced-form within-period response that combines (i) the own-price response for that period, (ii) cross-period substitution toward or away from other hours whose prices also change, and (iii) mechanical re-timing of flexible end uses (notably automated heat pump control). A useful way to think about our estimates is therefore as evidence of a high *within-day interhour elasticity of substitution*, reflecting a strong willingness/ability to shift usage from high-price to low-price hours rather than as evidence that overall electricity demand is highly price elastic.

Overall, households that also adopt the heat pump time-of-use tariff achieve meaningful reductions in their electricity bills. On average, these households save around £321 per year (18% of annual electricity expenditure), compared to if they were on the standard tariff from the same utility. Roughly one-quarter of these savings (around £76) arise from the tariff's structural price advantage — that is, the lower average price per kWh compared with the standard tariff even if consumption patterns do not change. The remaining three-quarters (about £245) stem from behavioral responses to the time-of-use price schedule, such as shifting heating and appliance use to cheaper off-peak periods.

Next, we explore how responses to the heat pump tariff vary across household and property characteristics. Households with higher annual electricity consumption, larger floor areas, or liv-

ing in higher-income areas exhibit larger absolute reductions in peak-period use, although these differences largely disappear when normalizing by each household's pre-period consumption profile. Energy efficiency bands show little systematic variation. We also find that households previously on time-of-use tariffs respond more strongly in the peak period, though they represent less than 10% of our sample and excluding them leaves our main estimates unchanged. Overall, these patterns indicate that while some groups shift more in absolute terms, the underlying behavioral response to the tariff is broadly similar across households. See Section A.4.10 for full heterogeneity plots.

### 2.2.3 Seasonal and Temperature Effects

We examine how the impact of the tariff on electricity consumption varies across the year and with outside temperature. In Figure A.21, we plot dynamic average treatment effects by calendar time across the four main periods of the day. We observe substantial load shifting into the morning and afternoon off-peak periods during colder months (October to March), with effects nearly disappearing in warmer months. Peak and "Other" period consumption fall correspondingly, indicating that off-peak increases are driven by substitution. Interestingly, the peak and "Other" reductions persist into summer, suggesting that some non-heating appliance usage is also being shifted.

Overall daily consumption increases moderately during winter but falls slightly in summer. These effects offset each other over the year, resulting in a near-zero average change. We hypothesize that in warmer months, the observed reduction in peak demand reflects some demand destruction (rather than pure shifting), whereas in colder months, some additional demand is created due to increased heating in off-peak hours.

A plausible mechanism is that households program their thermostats with higher setpoints during off-peak periods and lower setpoints during the peak period. Once set, this behavior is likely to persist over time and explains why shifting is strongest when the thermostat's setpoint is more likely to be binding — that is, when outdoor temperatures are low.

To further examine this hypothesis, we estimate the interaction between tariff adoption and daily average outside temperature, shown in Figure A.32. We control for household and day fixed effects and isolate the marginal impact of temperature after adoption. We find that the treatment effect nearly doubles during the coldest days relative to the average effect reported in Table A.9. Off-peak usage increases sharply when temperatures are below 14°C, consistent with heat demand being shifted from peak to cheaper hours. In the peak period, consumption decreases more steeply as temperature falls.

We do not observe a clear ceiling to shifting at very low temperatures, although very cold days (below 0°C) are rare in our data. This increasing responsiveness during colder weather may seem surprising, as one might expect households to resist lowering heating during peak hours when it is especially cold. However, since total heat demand is also higher in colder conditions (Section 2.1.3),

the absolute opportunity for shifting increases. Furthermore, heat pumps employ weather compensation: as the gap between indoor and outdoor temperatures widens, flow temperatures rise to maintain comfort. If households adopt lower peak-period thermostat setpoints under the heat pump tariff, this compensation mechanism amplifies the shifting effect.

Finally, outside temperature has little effect on consumption during non-heating periods, reinforcing the conclusion that most of the observed flexibility under the tariff relates to heating load.

### 2.3 Economic Implications of Heat Pump Adoption

Heat pump adoption generates two conceptually and economically distinct effects, each of which can be interpreted as concerning separate market imperfections. The first is an energy efficiency effect: heat pumps convert energy to useful indoor heat more efficiently than conventional gas boilers. This is reflected in substantial substitution from gas to electricity, with reductions in gas consumption over three times larger, in absolute terms, than the corresponding increase in electricity use. The second is a fuel-switching or congestion effect: by moving heating demand from energy sources whose timing is largely economically irrelevant (for example, due to pressurized gas systems) onto the electricity grid, households create new marginal costs that vary over time and are not fully captured by flat or weakly time-varying retail prices.

The first effect naturally raises the classic rebound question. If the implicit marginal cost of delivered heat falls, households may consume more heat, partially offsetting the engineering energy efficiency gains. In our setting, however, British electricity prices remain substantially higher per kWh than gas, meaning that the marginal cost of heating does not decline materially. For this reason, the scope for rebound is limited, and the observed energy efficiency gains are largely realized. Moreover, adopters in our sample may be early, higher-income households who already manage their heating to achieve desired comfort levels, further constraining potential rebound. The large gas reductions relative to electricity increases, combined with comparable marginal carbon intensities for the two fuels, imply that heat pump adoption delivers substantial carbon emissions reductions. This carbon abatement increases as the grid becomes less carbon-intensive in the future (see Section 3). This finding contrasts with much of the residential energy-efficiency literature (for example, Fowlie et al. 2018), where ex post savings have often fallen short of engineering projections. In our setting, it may be that the fuel switching impacts are mostly mechanical and do not interact strongly with behavioral adjustments. In addition, the absence of a substantial operating cost reduction may limit rebound. Both mechanisms would help explain why realized performance aligns closely with ex ante expectations. With that said, we acknowledge that we lack data from households' thermostats, so we cannot comment directly on comfort taking.

The second effect concerns grid congestion and temporal mispricing. Electrifying heating exposes demand to the structure of the electricity grid, where the marginal social cost of consumption varies across hours due to both congestion and emissions intensity. In this sense, electrified heat-

ing inherits a familiar second-best problem in electricity markets: even when upstream policies (such as carbon pricing) raise wholesale prices, distorted retail tariffs prevent consumers from efficiently adjusting demand across hours (Borenstein and Bushnell 2022*a*,*b*). Our results illustrate how time-of-use pricing can mitigate this problem: Figure A.22 shows that households on the heat pump tariff shift electricity demand toward off-peak hours (early morning and afternoon) while reducing consumption during peak periods, with little effect on total daily usage. This demonstrates that tariffs primarily correct intra-day pricing distortions and congestion externalities, rather than inducing large reductions in overall consumption.

Taken together, these two channels highlight distinct policy considerations. The engineering energy efficiency gains from electrification are substantial. However, without time-varying prices, electrified heating may impose higher congestion costs or emissions at peak times than necessary. Time-of-use pricing therefore emerges as a natural complement to heat pump adoption, allowing the energy efficiency gains from electrification to translate into welfare improvements on the power system.

# 3 Welfare Analysis

## 3.1 Marginal Value of Public Funds

We use the estimates of energy demand from heat pump adoption to evaluate the welfare effects of the UK's heat pump subsidy, the Boiler Upgrade Scheme. Our approach follows the marginal value of public funds (MVPF) framework of Hendren and Sprung-Keyser (2020), which compares the total social benefits of a policy to its net fiscal cost. In this framework, the reference point for overall economic efficiency is the marginal cost of public funds for raising the revenue for that subsidy, so that a given MVPF implies lower economic efficiency when the opportunity cost of public expenditures is higher. We account for the fiscal and environmental components of the subsidy alongside transfers to households who would have adopted regardless.

A key limitation of our design is that we cannot directly estimate the additionality rate. By this, we mean the share of subsidy recipients who were induced to adopt by the subsidy, as opposed to those who would have adopted without it. Relatedly, because eligibility for the subsidy is not randomized, our design does not identify an intention-to-treat effect of the subsidy itself; our estimates are effects of realized heat pump adoption, and moving from them to the welfare impact of the subsidy requires the additionality assumption we adopt below. We follow Hahn et al. (2026) in assuming that half of all recipients are *subsidy-induced* (those who adopt only because of the subsidy; referred to as "marginal" in the MVPF literature) and half are *non-additional* (those who would have adopted regardless; "inframarginal" in the MVPF literature). This is broadly consistent with empirical evidence from comparable residential energy efficiency programs (Boomhower and Davis 2014; Davis 2024; Hahn et al. 2026), though additionality rates vary substantially across

programs and this assumption represents a meaningful source of uncertainty in our welfare estimates.

Under a linear adoption demand curve, a subsidy-induced adopter's private cost of adopting is distributed uniformly between zero and the subsidy amount, so on average the subsidy splits evenly: half compensates the household's private cost of adopting, and half is retained as consumer surplus. Induced households therefore value the subsidy at roughly half its face value, while non-additional recipients, who would have adopted regardless, value it at the full amount as a pure transfer. Sensitivity to this assumption is shown in Table 3.

The social benefits comprise reductions in carbon dioxide equivalent ($CO_2e$) and local air pollutants arising from the observed fall in gas use and rise in electricity consumption. The fiscal and resource costs include the subsidy expenditure, the average installation cost, avoided operating costs on a counterfactual gas boiler, and associated changes in government revenue such as VAT on appliances and energy bills. Private operating benefits beyond fuel switching, such as potential savings from time-of-use electricity tariffs or other complementary technologies, are not modeled separately in the MVPF calculation. To the extent that such factors increase the private value of adopting a heat pump, they are implicitly captured in our reduced-form assumption about the share of subsidy-induced versus non-additional adopters.

Under a range of parameter choices, our estimated MVPF remains above unity across most specifications, falling marginally below 1 only under the lower, damage-based, carbon values, at the 3.5% rate. The estimate is sensitive to underlying assumptions, particularly the additionality of subsidy recipients and the chosen value of avoided carbon (see Table 3). We find that the average household reduces gas use by 8,744 kWh per year and increases electricity consumption by 2,551 kWh per year. Using UK Government emissions factors for end-use electricity consumption, which account for upstream efficiency losses in electricity generation and transmission, this corresponds to a reduction of approximately 1.16 tonnes of $CO_2e$ per household per year at 2024 Great Britain grid intensities, rising to around 1.74 tonnes per year by 2043 as the Great Britain electricity grid decarbonizes. Over an assumed 20-year lifetime of a heat pump, this implies a cumulative reduction of roughly 31 tonnes of $CO_2e$ per household.

Under the UK Government's approach, carbon values are derived by first specifying the quantity of $CO_2e$ reductions required to meet future emissions targets and then reading off the corresponding cost from the marginal abatement cost curve. This procedure yields a carbon value of around £287 per tonne of $CO_2e$ (in $£_{2023}$) (Department for Energy Security and Net Zero 2023*a*). This method differs from the approach used in some other countries, such as the United States, where the social cost of carbon is typically estimated using integrated assessment models that quantify the marginal damages from an additional tonne of $CO_2e$ (US Environmental Protection Agency 2023).

Combining these carbon valuations with the subsidy's fiscal cost, the average installation cost, and associated tax and spending effects, we estimate an average MVPF of 1.23 across the full £7,500

subsidy, driven primarily by avoided carbon emissions. This estimate should be interpreted with the assumptions above in mind (see Table 3), but it compares favorably with the estimated returns to other policies aimed at reducing $CO_2$e emissions (Hahn et al. 2026). Table 3 presents a sensitivity analysis illustrating how our estimates change under alternative assumptions about discount rates and the definition of costs and benefits. Section A.6.5 extends this to a joint sensitivity over the full range of additionality and social-cost-of-carbon assumptions. Because the marginal MVPF declines with subsidy size, it is higher at low subsidy levels: the first £1 of subsidy returns an MVPF of 1.65, falling to 1 at the optimal subsidy discussed below.

Further, we calculate an MVPF that accounts for dynamic efficiency effects from learning-by-doing. Evidence from other heating technologies suggests cost declines of roughly 14% for each doubling of cumulative installations (Weiss et al. 2009). Applying a similar learning rate to heat pumps increases the MVPF on the first £1 of subsidy from 1.65 to 2.51, reflecting the long-run social value of accelerated cost reduction and adoption. Figure 8 illustrates the components of the MVPF calculation, showing how social benefits, fiscal costs, and externalities interact to generate net welfare gains.

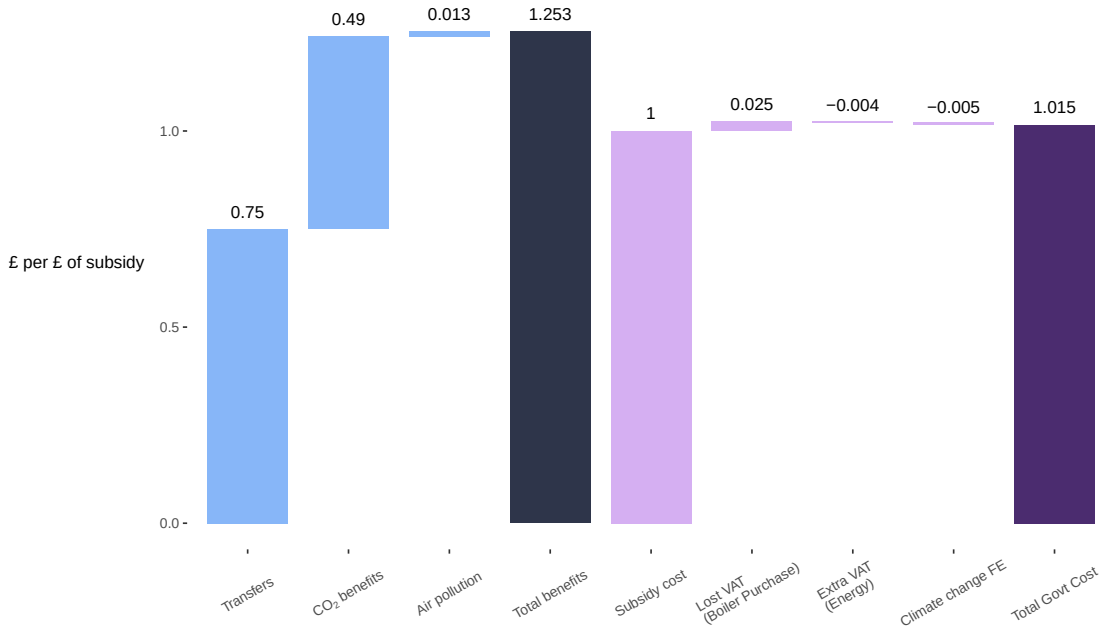
**Table 3:** MVPF and Other Measures of Cost-Effectiveness

| | | | MVPF | | | Cost per tonne | | |
|---|---|---|---|---|---|---|---|---|
| | Discount Rate | Additionality Rate | Average | First £ | First £ w/ LBD | Resource | Government | Social |
| MAC-based SCC | 3.5% | 50% | 1.23 | 1.65 | 2.51 | £86 | £487 | £329 |
| MAC-based SCC | 3.5% | 25% | 1.12 | 1.65 | 2.51 | £86 | £967 | £328 |
| IAM-based SCC | 2% | 50% | 1.03 | 1.37 | 2.23 | £84 | £488 | £328 |
| IAM-based SCC | 3.5% | 50% | 0.99 | 1.32 | 2.18 | £86 | £489 | £330 |

**Note:** There are multiple ways to evaluate the benefits and costs of the heat pump subsidy. We report the marginal value of public funds (MVPF) as a welfare measure, alongside resource, government, and social cost per tonne as measures of cost-effectiveness. Results vary by the appraisal discount rate (the UK Treasury recommends 3.5% (HM Treasury 2020)), and by the share of households that are subsidy-induced ("marginal" in the MVPF literature) versus non-additional ("inframarginal"). Carbon benefits are monetized either using UK marginal abatement cost-based carbon values (MAC), which assumes a social cost of carbon of approximately £287/tonne; or damage-based social cost of carbon estimates from integrated assessment models (IAM), which assumes a social cost of carbon of approximately £166/tonne ($\$_{2020}185$/tonne). In the IAM-based case, the social cost of carbon is taken from Rennert et al. (2022) and embeds a 2% near-term discount rate in its construction; in all cases, program costs and benefit streams are discounted at the appraisal rate shown in each row. Our preferred specification is the first row: a 3.5% appraisal discount rate and MAC-based social cost of carbon (as per UK Government guidance), and an assumption that 50% of households are subsidy-induced.

Based on these same assumptions, we also estimate an implied optimal subsidy (the point at which the marginal MVPF reaches 1, such that an additional £1 of subsidy generates no further societal benefit) of around £7,155, close to the current policy level, though this figure inherits the same sensitivities discussed above. If we include the learning-by-doing benefits in the calculation, the optimal subsidy rises to £16,551.

**Figure 8:** Marginal Value of Public Funds (MVPF) of the Boiler Upgrade Scheme




**Note:** We show the welfare benefits and costs of the subsidy for air-source heat pumps for the first row of Table 3 ("MAC-based", 3.5% discount rate, 50% of households as subsidy-induced). We show these benefits and costs per £7,500, which from October 2023 is the level of the subsidy offered to British households for adopting an air-source heat pump. In blue, we show benefits: transfers to both subsidy-induced and non-additional households receiving the subsidy, $CO_2$e abatement benefits from subsidy-induced households, and air quality benefits from subsidy-induced households. These total £9,397 per £7,500 subsidy (1.253, in navy). In light purple, we show the costs. First, the subsidy itself. Next, lost VAT from avoided boiler purchases from subsidy-induced households. Then, two increases in Government revenue that reduce the Government cost: extra VAT from increased energy expenditure and the "climate change fiscal externality" of tax revenue from avoided climate change. These total £7,615 per £7,500 (1.015, in dark purple). The average MVPF is then $1.253/1.015 = 1.23$.

It is worth emphasizing that our evaluation of heat pumps' impact is in the context of a subsidy that mandates full replacement of the previous heating system. In this context, the heat pumps lead to a nearly complete elimination of historical gas consumption. In other contexts, households might choose to keep their previous systems as back-up, which might dampen the fuel switching impacts, carbon abatement, and welfare benefits.

Households with poorly insulated homes, or those with a strong preference for rapid temperature recovery, may systematically prefer gas heating, while households with tighter building envelopes or stronger preferences for temperature stability may be more likely to adopt heat pumps. To the extent that adoption is selective on these dimensions, our estimates reflect the average treatment effect on the treated, namely outcomes for the observed population of adopters, and may not generalize to non-adopters, for whom the welfare effects of heat pump adoption could differ.

### 3.2 Heat Pump Impacts on Private Operating Costs

Separate from the MVPF calculation, we estimate impacts on households' operating costs. Based on our estimates and retail prices as of 15 June 2024, an average heat pump household spends about £10 less per year on energy than a comparable gas-heated home, using the utility's standard tariff's electricity and gas prices per kWh at that date (or £79 more per year using the Government's estimated retail energy costs in 2024). For context, we estimate that households paid approximately £2,100 per year for energy prior to heat pump adoption on the utility's standard tariff at those same prices. Against that benchmark, the £10/year implied operating cost saving at those prices reflects almost no change. Note that this is before any private cost savings from time-of-use pricing, which, as discussed in Section 2.2.2, reduce heat pump operating costs by £321/year.

Although we find that they use less total energy than the gas boilers they replace, the much higher retail unit price of electricity relative to gas offsets this efficiency gain. This in part reflects a broader mispricing of energy across fuels. Following Borenstein and Bushnell (2022*a*,*b*), fixed cost recovery in regulated utility markets raises retail prices above the private marginal cost of producing energy, an effect that is more pronounced for electricity because its infrastructure is more capital intensive and regulatory rate-making bundles a greater share of fixed costs into volumetric charges. At the same time, unpriced externalities mean that the private marginal cost falls below the marginal social cost, which can partially offset some of the cost recovery overpricing. A third mechanism further widens the price distortion: public-purpose programs. Certain policy programs can be funded through energy bills, as is the case in the UK for a number of policies. The resulting distortion weakens private incentives to electrify, even when electrification is socially beneficial. This pricing differential is a live policy issue in many countries. Recent UK policy changes, including the partial removal of electricity levies announced in November 2025 (HM Treasury 2025), as well as the European Commission's Action Plan for Affordable Energy (European Commission 2025), explicitly attempt to reduce the electricity–gas per-kWh price ratio, partly to support electrification and heat pump deployment.

Motivated by Borenstein and Bushnell (2022*a*,*b*), we can look at how households' operating costs would change under alternative pricing regimes that remove price distortions. We analyze a sequence of stylized pricing scenarios that progressively move retail energy prices toward marginal social cost. For each scenario, we explicitly reconstruct retail unit prices by identifying the relevant policy, network, and carbon cost components. We estimate the aggregate magnitude of these components using annual expenditures associated with major electricity subsidies, and convert these aggregates into per-kWh charges by allocating them over total national electricity or gas consumption from Department for Energy Security and Net Zero (2024).

Beyond the price baselines (1a)–(1d), our scenarios include: (2) removing part of legacy electricity policy costs from per-kWh charges in line with the UK Government's recent announcement (HM Treasury 2025); (3) removing all explicit policy costs from electricity unit prices; (4) remov-

ing volumetric recovery of electricity network costs[4]; (5) applying carbon pricing to household gas consumption, which is currently exempt; and (6) pricing carbon in both electricity and gas at the UK Government's central social cost of carbon. These scenarios are not intended as literal policy proposals, but rather as transparent benchmarks designed to quantify the magnitude of existing pricing distortions.[5]

**Table 4:** Heat Pump Operating Cost Differences under Alternative Pricing Regimes

| Scenario | Heat pump penalty (+) / saving (–), £ per year | Gas price (£/kWh) | Electricity price (£/kWh) |
|---|---|---|---|
| (1a) Govt estimated prices | £186 | £0.0816 | £0.3528 |
| (1b) Govt estimated prices, no gas standing charge | £79 | £0.0816 | £0.3528 |
| (1c) Utility prices | £97 | £0.0604 | £0.2450 |
| (1d) Utility prices, no gas standing charge | –£10 | £0.0604 | £0.2450 |
| (2) Partial electricity policy costs reduction | –£49 | £0.0589 | £0.2245 |
| (3) Full removal of electricity policy costs | –£106 | £0.0589 | £0.2022 |
| (4) Remove electricity policy costs + network charges | –£227 | £0.0589 | £0.1547 |
| (5) Scenario 4 + gas priced under UK ETS | –£326 | £0.0701 | £0.1547 |
| (6) MSC: Scenario 4 + both fuels priced at SCC | –£614 | £0.1170 | £0.2026 |

**Note:** Entries show the change in annual operating costs for a heat pump household relative to a gas boiler. Positive values indicate a higher cost; negative values indicate savings. Calculations assume fixed annual energy use of 2,551 kWh electricity and -8,744 kWh gas. Scenarios (1a)–(1d) vary retail price assumptions and treatment of the gas standing charge (£107 per year), with Scenario (1d) used as the main benchmark. Scenarios (2) and (3) progressively remove policy costs, which mostly fall on electricity. Scenario (4) removes *network* costs from unit rates, as well. Scenario (5) includes all of these changes, but introduces carbon pricing on gas (which is currently exempt, when used for home heating). Scenario (6) brings both fuels to their marginal social cost (MSC), where electricity unit rates have policy and network costs removed, and both fuels are priced at the UK Government's official social cost of carbon (SCC). All scenarios are stylized counterfactuals intended to illustrate the magnitude of pricing effects rather than specific policy proposals. For context, we estimate that households paid approximately £2,100 per year for energy prior to heat pump adoption on the utility's current standard tariff. Note that this is before any private cost savings from time-of-use pricing, which, as discussed in Section 2.2.2, reduce heat pump operating costs by £321/year. More detailed notes are available in Section A.6.6.

By rebalancing electricity and gas prices toward their marginal social costs, removing upward distortions from fixed cost recovery and public-purpose programs while internalizing externalities, these scenarios quantify the extent to which the observed operating-cost differential reflects pricing institutions rather than the underlying efficiency of heat pump technology. Table 4 reports the implied change in annual household heat pump operating costs under each scenario, relative to a gas boiler, holding usage fixed and abstracting from installation costs. As electricity "headwinds" are progressively removed and gas prices move closer to their marginal social cost, the operating cost savings from heat pumps increase.

[4]The question of how to recover electricity network costs is itself an important economic consideration. For example, McRae and Wolak (2024) argue for greater reliance on customer-specific fixed charges when recovering network costs.

[5]We note two simplifications. First, Scenario 6 prices only carbon (via the UK Government's social cost of carbon) and does not additionally price local air pollution externalities (e.g., from gas combustion or marginal electricity generation), so it understates a fully comprehensive marginal-social-cost benchmark. Second, we hold standing (fixed) charges constant across scenarios even as we vary volumetric unit prices; in practice, a regulator recovering fixed network costs would likely adjust standing charges to preserve revenue neutrality. This is particularly relevant for the gas standing-charge disconnection scenarios (1b, 1d), where at scale, other gas customers would face higher standing charges to recover the network costs no longer collected from disconnecting households. We view both simplifications as second-order relative to the main pricing distortions discussed.

In presenting these counterfactual operating cost impacts, we assume that changes in electricity and gas prices per kWh do not induce changes in household energy consumption. Estimated changes in electricity use (2,551 kWh) and gas use (-8,744 kWh) are therefore held fixed across all pricing scenarios. This is a strong assumption, since households may adjust usage in response to price changes. Our objective, however, is not to model behavioral demand responses under alternative policy regimes, but to provide transparent, stylized calculations that isolate the mechanical effect of different pricing structures on relative operating costs. The scenarios should therefore be interpreted as accounting exercises that quantify the magnitude of existing pricing wedges and illustrate how policy "headwinds" and "tailwinds" shape the private economics of heat pump adoption.

Finally, note that the analysis uses 2024 operating costs and does not model how these pricing regimes would evolve over a 20-year equipment lifetime. As the electricity grid continues to decarbonize, consistent carbon pricing would likely reduce the carbon cost component of electricity relative to gas, strengthening the operating cost advantage of heat pumps over time, especially in Scenarios 5 and 6. Even in the static comparisons presented here, however, the results highlight an important contrast: while the baseline operating cost difference is modest relative to overall household energy expenditure, the policy wedges embedded in retail prices are economically meaningful. Removing electricity policy and network costs and introducing carbon pricing on gas materially shift the relative cost comparison, making heat pumps more clearly cost saving under pricing regimes that better align retail prices with marginal social costs.

An implication of this analysis that we do not fully explore is its interaction with the welfare and subsidy calculations in Section 3. Heat pump adoption in Great Britain remains low relative to the stock of technically suitable homes, and our MVPF and optimal-subsidy estimates are calibrated to this setting, characterizing adoption behavior at the margin under current retail prices, where the £2,963 incremental cost of a heat pump over a gas boiler (Section A.6) is effectively never recovered (Scenario 1d). A policy-relevant question raised by fuel mispricing is how large a subsidy, or how much price reform, would be needed to move most households with suitable homes to voluntarily choose a heat pump over a gas boiler. Table 4 suggests that even under our most aggressive re-pricing scenario (Scenario 6), where operating costs fall by approximately £600 per year, the implied payback period falls to about five years, a substantial improvement over the status quo, and plausibly fast enough to affect adoption decisions for a meaningful share of households.

We therefore view current mispricing as one plausible contributor to slow diffusion, and correcting it as likely to reduce, and even potentially eliminate, the subsidy required for widespread adoption among suitable households. This is not a coincidence but a substitution between instruments: Scenario 6 internalizes the same carbon externality that drives our MVPF above 1, so pricing carbon correctly and subsidizing adoption are alternative corrections for one distortion. Any residual case for a subsidy would then rest on the market failures that carbon pricing does not address, such as learning-by-doing spillovers and credit constraints. That said, re-pricing would also affect

government revenue directly and independent of any adoption response. For instance, applying carbon pricing to gas would raise revenue, while removing policy costs from electricity bills would lower it. A full accounting would need to net these effects against any fiscal savings from a lower required subsidy. Quantifying these interactions properly would require adoption elasticities with respect to payback periods that are not currently available in the literature or estimable from our data, so we leave a fuller treatment of pricing and subsidy design for mass adoption to future work.

# 4 Conclusion

Our study shows that heat pump adoption induces substantial substitution away from fossil fuels: household gas consumption falls sharply, while electricity use rises by around 30% as much in kWh terms. These relative effect sizes reflect the higher energy efficiency of heat pumps versus gas boilers. The primary policy relevance of this fuel switch lies not in aggregate energy savings per se, but in the resulting emissions reductions. Under the current UK electricity generation mix, carbon emissions from household energy consumption fall by approximately 39%. Under projected grid decarbonization pathways (UK Government 2024), cumulative emissions reductions from heat pump adoption rise to around 72% over 20 years.

Economically, these effects operate in two related ways. First, heat pumps deliver energy efficiency gains in converting energy into useful heat. In contrast to much of the prior energy-efficiency literature, we find little evidence of rebound, which means the engineering energy efficiency gains translate closely into realized energy and carbon savings. Second, electrification shifts heating demand onto the electricity grid, where the marginal social cost of consumption varies over time. This creates a congestion and temporal mispricing problem that is largely absent under gas heating.

We show that time-of-use pricing can meaningfully mitigate this distortion. Households on the heat pump tariff reallocate demand away from peak periods and toward off-peak hours, with little change in total daily consumption. The primary economic role of such tariffs is therefore not to reduce overall electricity use, but to improve the alignment between household demand and underlying system conditions, which could ease network reinforcement needs and lower system costs associated with electrification, as suggested in energy system modeling (Franken et al. 2025). In this way, tariff design emerges as a natural complement to electrification.

These findings come with a caveat about external validity. Heat pump adopters in our sample are likely those for whom heat pumps are most desirable or well suited, so the observed effects need not reflect population treatment effects; we expect them to generalize reasonably well nonetheless, since the response is largely mechanical and tied to outdoor temperature rather than to idiosyncratic traits of early adopters. The heat pump tariff results are more sensitive to selection into treatment by structural winners and low-shifting-cost households: our sample skews toward larger homes and, we suspect, more technologically engaged households, which likely contributes to the large observed load-shifting response. As adoption spreads to smaller homes with occupants

less familiar with automation, we would expect smaller absolute effects, though still meaningful ones relative to heat pump and home size. Estimating these effects in a representative population, through experimental variation in tariff assignment, or in settings where adoption is less selected, would be a valuable direction for future work.

Our analysis also clarifies the role of retail energy pricing in shaping private incentives. Under current retail prices, heat pumps do not meaningfully reduce household operating costs. Electricity prices embed significant policy and network costs through volumetric charges, while household gas prices largely exclude climate externalities. This pricing structure weakens private incentives to electrify, even when electrification is socially beneficial. When retail prices move closer to marginal social cost, through rebalancing electricity levies or pricing carbon consistently across fuels, the private operating cost savings from heat pumps increase substantially. In this sense, current pricing creates "headwinds" for electrification (Borenstein and Bushnell 2022*b*) that are not rooted in the underlying energy efficiency of the technology. However, even under current retail prices, the addition of a time-of-use tariff can bring heat pump operating costs below average boiler operating costs.

From a welfare perspective, our calculations suggest the UK heat pump subsidy is welfare-enhancing under a range of plausible assumptions, largely driven by significant carbon savings. As discussed in Section 3, this conclusion rests on assumptions, including the additionality of subsidy recipients and the social cost of carbon used, that are each subject to meaningful uncertainty. The societal value of heat pump adoption would increase further if technology costs fall, including from learning by doing seen in the deployment of other technologies, and if the carbon intensity of electricity generation falls faster than expected.

Overall, heat pumps should be understood as a decarbonization technology that combines energy efficiency gains with fuel switching. Their social value would be enhanced by complementary policies: electricity and gas pricing regimes to align private incentives with marginal social costs, and time-of-use pricing to mitigate the impact of electrified heating on system costs.

## References


Allcott, H. and Greenstone, M. (2017), Measuring the welfare effects of residential energy efficiency programs, Working Paper 23386, National Bureau of Economic Research.

Anderson, S. and Kirkpatrick, A. J. (2024), 'Distributional consequences of policies for electric heat conversion', *Journal of the Association of Environmental and Resource Economists* **11**(S1), S9–S40.

Bailey, M. R., Brown, D. P., Myers, E., Shaffer, B. and Wolak, F. A. (2025), 'Electric vehicles and the energy transition: Unintended consequences of time-of-use pricing', *American Economic Review: Insights* **7**(4), 550–566.

Blonz, J., Palmer, K., Wichman, C. J. and Wietelman, D. C. (2025), 'Smart thermostats, automation, and time-varying prices', *American Economic Journal: Applied Economics* **17**(1), 90–125.

Bollinger, B. K. and Hartmann, W. R. (2020), 'Information vs. automation and implications for dynamic pricing', *Management Science* **66**(1), 290–314.

Boomhower, J. and Davis, L. W. (2014), 'A credible approach for measuring inframarginal participation in energy efficiency programs', *Journal of Public Economics* **113**, 67–79.

Borenstein, S. (2007), 'Wealth transfers among large customers from implementing real-time retail electricity pricing', *The Energy Journal* **28**(2), 131–150.

Borenstein, S. and Bushnell, J. B. (2022*a*), 'Do two electricity pricing wrongs make a right? Cost recovery, externalities, and efficiency', *American Economic Journal: Economic Policy* **14**(4), 80–110.

Borenstein, S. and Bushnell, J. B. (2022*b*), 'Headwinds and tailwinds: Implications of inefficient retail energy pricing for energy substitution', *Environmental and Energy Policy and the Economy* **3**(1), 37–70.

Brandon, A., Clapp, C. M., List, J. A., Metcalfe, R. D. and Price, M. (2022), The human perils of scaling smart technologies: Evidence from field experiments, Working Paper 30482, National Bureau of Economic Research.

Callaway, B. and Sant'Anna, P. H. C. (2021), 'Difference-in-differences with multiple time periods', *Journal of Econometrics* **225**(2), 200–230.

Christensen, P., Francisco, P., Myers, E. and Souza, M. (2023), 'Decomposing the wedge between projected and realized returns in energy efficiency programs', *The Review of Economics and Statistics* **105**(4), 798–817.

Climate Change Committee (2025), The Seventh Carbon Budget: Advice to Parliament, Technical report, Climate Change Committee, London.

Davis, L. W. (2024), 'The economic determinants of heat pump adoption', *Environmental and Energy Policy and the Economy* **5**(1), 162–199.

Department for Energy Security and Net Zero (2023*a*), 'Green Book supplementary guidance: Valuation of energy use and greenhouse gas emissions for appraisal'.

Department for Energy Security and Net Zero (2023*b*), 'Greenhouse gas reporting: Conversion factors 2023'.

Department for Energy Security and Net Zero (2024), 'Digest of UK Energy Statistics (DUKES) 2024'.

Department of Energy and Climate Change (2009), In-situ monitoring of efficiencies of condensing boilers and use of secondary heating: Final report, Technical report, Department of Energy and Climate Change, London, United Kingdom. Research carried out by GASTEC at CRE Ltd, AECOM, and EA Technology; commissioned via the Energy Saving Trust.

Electric Power Research Institute (2018), US-REGEN model documentation, Technical report, Electric Power Research Institute, Palo Alto, CA.

European Commission (2025), 'Action Plan for Affordable Energy: Unlocking the true value of our Energy Union to secure affordable, efficient and clean energy for all Europeans', Communication, European Commission. COM(2025)79.

Fowlie, M., Greenstone, M. and Wolfram, C. (2018), 'Do energy efficiency investments deliver? Evidence from the Weatherization Assistance Program', *The Quarterly Journal of Economics* **133**(3), 1597–1644.

Franken, L., Hackett, A., Lizana, J., Riepin, I., Jenkinson, R., Lyden, A., Yu, L. and Friedrich, D. (2025), 'Power system benefits of simultaneous domestic transport and heating demand flexibility in Great Britain's energy transition', *Applied Energy* **377**, 124522.

Freyaldenhoven, S., Hansen, C. and Shapiro, J. M. (2019), 'Pre-event trends in the panel event-study design', *American Economic Review* **109**(9), 3307–3338.

Gillan, J. M. (2017), Dynamic pricing, attention, and automation: Evidence from a field experiment in electricity consumption, Working Paper 284, Energy Institute at Haas, University of California, Berkeley.

Goodman-Bacon, A. (2021), 'Difference-in-differences with variation in treatment timing', *Journal of Econometrics* **225**(2), 254–277.

Hahn, R. W., Hendren, N., Metcalfe, R. D. and Sprung-Keyser, B. (2026), 'A welfare analysis of policies impacting climate change', *American Economic Review* **116**(7), 2368–2421.

Haldane, T. G. (1930), 'The heat pump—an economical method of producing low-grade heat from electricity', *Journal of the Institution of Electrical Engineers* **68**(402), 666–675.

Hendren, N. and Sprung-Keyser, B. (2020), 'A unified welfare analysis of government policies', *The Quarterly Journal of Economics* **135**(3), 1209–1318.

HM Treasury (2020), 'The Green Book: Appraisal and evaluation in central government'.

HM Treasury (2025), 'Budget 2025'. Accessed 16 February 2026.

Imelda, Fripp, M. and Roberts, M. J. (2024), 'Real-time pricing and the cost of clean power', *American Economic Journal: Economic Policy* **16**(4), 100–141.

International Energy Agency (2021), 'Net Zero by 2050: A roadmap for the global energy sector'.

International Energy Agency (2023), 'Renewables 2023: Heat'.

IPCC (2022), 'Climate change 2022: Mitigation of climate change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change'.

Ito, K. (2014), 'Do consumers respond to marginal or average price? Evidence from nonlinear electricity pricing', *American Economic Review* **104**(2), 537–563.

Ito, K., Ida, T. and Tanaka, M. (2023), 'Selection on welfare gains: Experimental evidence from electricity plan choice', *American Economic Review* **113**(11), 2937–2973.

Jessoe, K. and Rapson, D. (2014), 'Knowledge is (less) power: Experimental evidence from residential energy use', *American Economic Review* **104**(4), 1417–1438.

McRae, S. D. and Wolak, F. A. (2024), Electricity pricing and the energy transition for residential and non-residential consumers, Research Department Working Paper 2266, CAF Development Bank of Latin America, Caracas.

Microgeneration Certification Scheme (MCS) (2024), 'MCS data dashboard'. Accessed 24 July 2024.

Octopus Energy (2024), 'SVT April 2024 pricing rates'.

Rennert, K., Errickson, F., Prest, B. C., Rennels, L., Newell, R. G., Pizer, W., Kingdon, C., Wingenroth, J., Cooke, R., Parthum, B., Smith, D., Cromar, K., Diaz, D., Moore, F. C., Müller, U. K., Plevin, R. J., Raftery, A. E., Ševčíková, H., Sheets, H., Stock, J. H., Tan, T., Watson, M., Wong, T. E. and Anthoff, D. (2022), 'Comprehensive evidence implies a higher social cost of $CO_2$', *Nature* **610**(7933), 687–692.

Rosenow, J., Lea, T. and Boni, G. (2026), 'Bridging the efficiency divide: Open-source insights into UK heat pump performance gaps', *Energy and Buildings* **352**, 116785.

Roth, J. (2022), 'Pretest with caution: Event-study estimates after testing for parallel trends', *American Economic Review: Insights* **4**(3), 305–322.

Sergici, S., Ramakrishnan, A., Kavlak, G., Bigelow, A. and Diehl, M. (2023), Heat pump–friendly cost-based rate designs, Technical report, Energy Systems Integration Group.

UK Government (2024), 'Data tables 1–19 from the Green Book: Appraisal and evaluation in central government'.

U.S. Department of Energy (2023), 'Purchasing energy-efficient residential gas boilers', Federal Energy Management Program, U.S. Department of Energy. Accessed 17 May 2025.

US Environmental Protection Agency (2023), Report on the social cost of greenhouse gases: Estimates incorporating recent scientific advances, Technical report, US Environmental Protection Agency, Washington, DC. Final report.

Watson, S. D., Crawley, J., Lomas, K. J. and Buswell, R. A. (2023), 'Predicting future GB heat pump electricity demand', *Energy and Buildings* **286**, 112917.

Weiss, M., Dittmar, L., Junginger, M., Patel, M. K. and Blok, K. (2009), 'Market diffusion, technological learning, and cost-benefit dynamics of condensing gas boilers in the Netherlands', *Energy Policy* **37**(8), 2962–2976.

Wolak, F. A. (2011), 'Do residential customers respond to hourly prices? Evidence from a dynamic pricing experiment', *American Economic Review* **101**(3), 83–87.